

Multi-dimensional normalization for powder inelastic neutron experiments

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September 2026

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Neutron Scattering Division

**MULTI-DIMENSIONAL NORMALIZATION
FOR POWDER INELASTIC NEUTRON EXPERIMENTS**

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September 2026

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managed by
UT-BATTELLE LLC
for the
US DEPARTMENT OF ENERGY
under contract DE-AC05-00OR22725

CONTENTS

LIST OF FIGURES	iv
LIST OF ABBREVIATIONS	v
ABSTRACT	1
1. THEORY	1
2. IMPLEMENTATION DETAILS	3
2.1 CALCULATING INTERSECTIONS	3
2.2 ASSIGNING INTERSECTIONS TO THE CORRECT BIN	3
2.3 NORMALIZATION CONTRIBUTION	4
3. ALGORITHM STEPS	5
3.1 NOTES:	5
4. REFERENCES	6

LIST OF FIGURES

Figure 1.	Detector trajectories in a powder inelastic measurement on a direct geometry instrument, for an incident energy of 20 meV.	2
Figure 2.	Detector trajectories in a powder inelastic measurement on an indirect geometry instrument, for a fixed final energy of 5 meV.	4

LIST OF ABBREVIATIONS

DGS	Direct Geometry Spectroscopy
IGS	Indirect Geometry Spectroscopy
MDE	Multi-Dimensional Event
MDH	Multi-Dimensional Histogram
ORNL	Oak Ridge National Laboratory

ABSTRACT

Routine event-based processing is one of the strengths of neutron scattering experiments at Oak Ridge National Laboratory (ORNL) (Peterson et al. 2015, Granroth et al. 2018, Peterson et al. 2018). Multi-Dimensional Event (MDE) workflows are already well-established for single-crystal Direct Geometry Spectroscopy (DGS) and diffraction experiments. This report demonstrates how the same formalism can be extended to powder measurements.

1. THEORY

The purpose of a Direct Geometry Spectroscopy (DGS) or Indirect Geometry Spectroscopy (IGS) neutron experiment is to measure the double differential cross section, or equivalently the dynamic structure factor (Lovesey 1984:

$$\frac{d^2\sigma}{dEd\Omega} = N_S \frac{k_f}{k_i} |b(\mathbf{Q})|^2 S(\mathbf{Q}, \Delta E) \quad (1)$$

Here N_S is the number of scatterers, k_i, k_f are the incident/scattered neutron momentum, b is the scattering length, $\mathbf{Q}$ and ΔE are the momentum and energy transfers to the sample.

Measurements of powder samples on DGS and IGS instruments are typically stored as histograms with uniform binning along $|\mathbf{Q}|$ and ΔE . For each two-dimensional bin, the differential cross section can be written as (Michels-Clark et al. 2016, Savici et al. 2022)

$$\frac{d^2\sigma}{dEd\Omega} = \frac{\sum_j N_j}{\sum_l \Phi_l dE_l d\Omega_l} \quad (2)$$

N represents the number of neutrons, Φ is the incident flux, $d\Omega$ is the solid angle of a detector, and dE is an energy step.

During the experiment, several measurements and several detectors in each measurement can contribute to this bin. One needs to account separately for the neutron counts and separately for the normalization factors. In equation 2, the sum over index j takes all the neutrons scattered in the bin, independent on which measurement/detector it is coming from. The sum over index l in the denominator is a little more complicated. In each measurement (run), we are interested in finding the contributions from each detector. For DGS, if all measurements are performed with the same incident energy, the flux Φ_l is proportional (and can be replaced) by the proton charge of the run. For IGS, the calculation of the flux is based on the amount of incident beam that can contribute to a particular bin for a particular detector. This is similar to the flux calculation for single crystal experiments, as shown in Savici et al. 2022. The exact solid angle $d\Omega_l$ for a pixel is unknown, but can be measured using the counts in that particular detector when the scatterer is isotropic (Vanadium). The remaining term, dE_l , is the length along energy transfer of the trajectory in the given bin.

To better understand this, we depict trajectories in the DGS case for several detectors in Figure 1.

For each detector, at a scattering angle θ , we can compute this from the conservation of energy and momentum:

$$\Delta E = E_i - E_f \quad (3)$$

$$|\mathbf{Q}|^2 = k_i^2 + k_f^2 - 2k_i k_f \cos \theta, \quad (4)$$

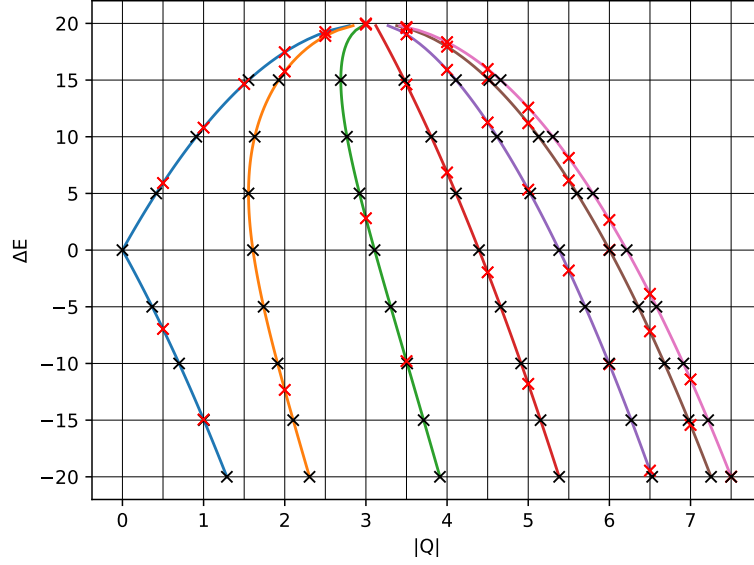


Figure 1. Detector trajectories in a powder inelastic measurement on a direct geometry instrument, for an incident energy of 20 meV. Colored curves correspond to detectors at 0, 30, 60, 90, 120, 150, 180°.

The grid delimits bins with dimensions $0.5\text{\AA}^{-1}$ along $|Q|$, and 5 meV along ΔE . Intersections of trajectories with constant ΔE lines are marked with black symbols. Intersections of trajectories with constant $|Q|$ lines are marked with red symbols.

with $E_{i,f}$ being the incident and scattered neutron energies. In addition, using m for the neutron mass,

$$E_{i,f} = \frac{\hbar^2 k_{i,f}^2}{2m} \quad (5)$$

For a particular detector, the trajectory is a curve in $(|Q|, \Delta E)$ space. For all ΔE values between some minimum and maximum, one calculates k_f from equations 3 and 5, then use it in equation 4.

If a trajectory intersects a bin in the histogram, its contribution is given by the length along the energy transfer direction that is contained within that bin. Calculating intersections with a set of parallel planes along momentum and energy transfer was already described before (Savici et al. 2022) for single crystal experiments, taking advantage of the linearity of the trajectory in terms of final momentum. For powder-averaged measurements, the trajectory becomes non-linear, as illustrated in equations 3 - 5, and shown in Figure 1. In the next section we show how to perform this calculation.

2. IMPLEMENTATION DETAILS

Particular implementation details are described in terms of the Mantid software (Arnold et al. 2014) in use at Oak Ridge National Laboratory (ORNL), but the formalism is entirely general. The workflow starts with a set of neutron events, labeled by momentum and energy transfer. The storage container in Mantid is an Multi-Dimensional Event (MDE) workspace. It contains information about instrument geometry and various experimental parameters. At minimum, the procedure below requires knowledge of incident energy E_i , and the minimum and maximum energy transfer that was used for each detector. Similar to the single crystal case, the first step is binning the neutron events into a Multi-Dimensional Histogram (MDH). One needs to create on the same grid a histogram with the normalization factors (denominator in the equation 2). The final, normalized intensity is the ratio of these histograms.

2.1 CALCULATING INTERSECTIONS

For each detector (or detector group) one calculates intersections with constant energy transfer lines in exactly the same way as we calculate the trajectories, but uses only energy transfers that are bin edges. In addition, one needs to consider the points that are at the minimum/maximum available energy transfer. These might not be on a bin boundary.

In the case of DGS, the intersections with constant Q lines yield final momentum values from equation 4.

$$k_f^\pm = k_i \cos \theta \pm \sqrt{Q^2 - k_i^2 \sin^2 \theta} \quad (6)$$

For IGS one calculates the incident momentum using

$$k_i^\pm = k_f \cos \theta \pm \sqrt{Q^2 - k_f^2 \sin^2 \theta} \quad (7)$$

The calculation for IGS case is shown in Figure 2.

Note that if the discriminant is negative, there are no solutions. Also, if the calculated ΔE for any of these values is outside the minimum/maximum range of energy transfers, then these values should not be stored. Negative $k_{i,f}$ solutions are not physical.

Once all intersections and end points are calculated for a particular detector, they need to be sorted by ΔE .

2.2 ASSIGNING INTERSECTIONS TO THE CORRECT BIN

When processing data, including the transformation to $|Q|$ and ΔE , each detector trajectory covers a limited range in energy transfer. This can be stored as *MDNorm_low* and *MDNorm_high* logs in the MDE workspace. For all intersections, ΔE must be greater than both *MDNorm_low* for the detector and the lowest ΔE for the grid. Similarly, for the upper bound, all intersections have to be less than *MDNorm_high* for the detector and the highest ΔE for the grid.

Consecutive intersections have different energy transfers, unless they have the same $|Q|$ value as well. If they differ in the $|Q|$ value one should assign intensity to the box that the center between these intersections falls into. corresponding to $(E_i + E_{i+1})/2$ and $(|Q|_i + |Q|_{i+1})/2$. Because the trajectories are non-linear, consecutive intersection points may have identical $|Q|$ values. In equation 4, $|Q|^2$ is a convex quadratic function of either k_i or k_f . Consequently, any momentum value lying between two solutions corresponding to the same original $|Q|^2$ will yield a smaller momentum transfer. Therefore, the intensity should be assigned to the bin on the lower- $|Q|$ side.

No intensity should be assigned for any $|Q|$ outside the minimum/maximum momentum transfer of the grid.

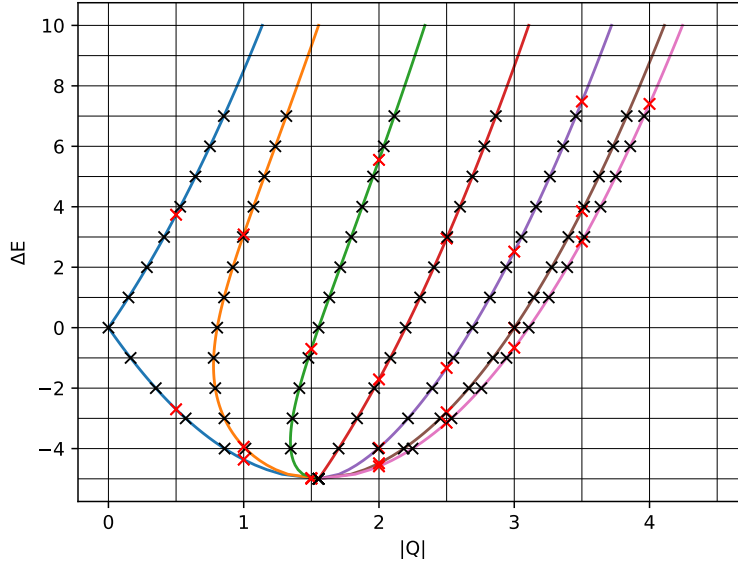


Figure 2. Detector trajectories in a powder inelastic measurement on a indirect geometry instrument, for a fixed final energy of 5 meV. Colored curves correspond to detectors at 0, 30, 60, 90, 120, 150, 180°. The grid delimits bins with dimensions $0.5\text{\AA}^{-1}$ along $|Q|$, and 1 meV along ΔE . Intersections of trajectories with constant ΔE lines are marked with black symbols. Intersections of trajectories with constant $|Q|$ lines are marked with red symbols.

2.3 NORMALIZATION CONTRIBUTION

The denominator of equation 2 is a normalization factor for the neutron counts. For each detector (or detector group), and for every run, the number of events scattered in a particular bin is proportional to the solid angle, the incident neutron flux that can contribute to that particular bin, and the length of the trajectory along the energy transfer direction.

As mentioned in previous sections, the solid angle is proportional to the neutron counts measurements of incoherent scatterers.

For DGS measurements all incident neutrons can contribute to scattering. If one uses a single energy/ configuration of the incident beam, the flux is proportional to the integrated proton charge. This approach provides better statistical precision than relying on integrated monitor counts, especially for short runs. However, it does not account for different incident flux as function of selected fixed energy/ configuration of incident beam.

For IGS measurements, due to conservation of momentum and energy, not all incident neutrons can contribute to scattering in a particular histogram bin. We have identified in section 2.1 the intersection of the detector trajectories with histogram bins. Neutrons with incident energies between consecutive intersections contribute to the histogram bins identified in section 2.2.

3. ALGORITHM STEPS

Mantid MDE workspaces can store information from separate measurements. Instrument geometry and experimental sample logs are stored in "experiment info" structures.

- The first step is validation. If one uses grouped detectors for the input MDE workspace, one must use the same grouping for the incoherent (Vanadium) workspace. The Vanadium workspace must be integrated into a single bin corresponding to the elastic line.
- Histogram the events.
- Create a similar grid for normalization, initialized to 0.
- Additional dimensions should be processed similar to the MDNorm algorithm in Mantid. We can use temperature as an example. For every experiment info check first if the temperature log value falls into one of the bins along the temperature grid. That particular experiment info will contribute only to one temperature bin.
- For every experiment info, loop over detectors or detector groups, calculate intersections. Discard any negative k_f values, or any values that fall outside the histogram edges. For each consecutive intersections, assign the intensity to the corresponding box (inside the histogram). The intensity to be added is equal to the proton charge in the experiment info, multiplied with the vanadium intensity for that particular detector group, and the energy transfer difference between intersections. For the case of IGS, the flux is proton charge multiplied with integrated flux spectrum between intersections.

3.1 NOTES:

- The Mantid implementation of the algorithm relies on the MDE workspace storing correctly the detector grouping. This needs to be persisted during saving/loading from files.
- For IGS, the energy dependence of the incident flux can be found from a monitor spectrum. It is important to correct it for the energy response of the monitor itself. Similarly, for the DGS one needs to correct the counts for the energy dependence of the detectors.
- Since DGS instruments at ORNL use a large number of finely pixelated detectors, one can group detectors with similar scattering angle in order to decrease the number of calculations.
- For DGS one can use multiple incident energies, as seen in super-resolution measurements (Islam et al. 2019). In this case one needs to account for incident beam flux energy dependence.

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